

M-Tech Production & Marketing

Internship Program 2026

TASK 3 REPORT

Submitted By:	Ayesha Siddiqua
Department:	Cyber Security
Submitted To:	Mohazin Zahid
Supervisor:	Mohazin Zahid
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Introduction

This report presents the work completed for Task 4 of the M-Tech Production & Marketing Cyber Security Internship Program, Batch 2026. Task 3 falls under the Networking topic area and focuses on building practical familiarity with common network ports, the services that run on them, and the security risks associated with each.

M-Tech Production & Marketing is a forward-looking, FBR-registered company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. As part of the structured 9-Phase internship roadmap, this task builds on earlier orientation and foundational sessions by moving into hands-on, tool-based technical work in the Networking domain.

Objectives

- To research and document 20 commonly used network ports and their associated services.
- To understand the security risks and common attack vectors linked to each port/service.
- To gain hands-on experience using nmap to identify open ports on a local system.
- To produce a concise, professional reference document (cheat-sheet) summarizing the findings.

Task Description

Task 3 was assigned under the Networking topic of the internship roadmap, at Beginner level. The task required building a cheat-sheet of 20 common ports, documenting their associated service and security risk, using research and an nmap scan restricted to localhost only, and delivering the result as a PDF.

The following table summarizes the 20 ports researched, along with their protocol, service, common use, and associated security risk:

Port	Protocol	Service	Common Use	Security Risk
20/21	TCP	FTP (Data/Control)	File transfer	Unencrypted creds & data; sniffable, brute-forceable
22	TCP	SSH	Secure remote login	Target for brute-force; weak keys/passwords risky
23	TCP	Telnet	Remote login (legacy)	Fully unencrypted; credentials sent in plaintext
25	TCP	SMTP	Mail relay/sending	Open relay abuse, spoofing, spam relaying
53	TCP/UDP	DNS	Name resolution	Cache poisoning, DNS amplification DDoS, exfiltration
67/68	UDP	DHCP	IP address assignment	Rogue DHCP servers, spoofing, DoS
69	UDP	TFTP	Simple file transfer	No auth or encryption; easy data theft/tampering
80	TCP	HTTP	Web traffic	Unencrypted; vulnerable to MITM and data interception
110	TCP	POP3	Mail retrieval	Plaintext auth by default; credential exposure
111	TCP/UDP	RPCbind/Portmapper	RPC service mapping	Info disclosure, used in DDoS reflection attacks
123	UDP	NTP	Time synchronization	NTP amplification DDoS attacks
135	TCP	MS RPC	Windows RPC services	Common exploit target (e.g., worms), info leakage
137-139	TCP/UDP	NetBIOS	Windows file/printer sharing	Info disclosure, legacy SMB attacks
143	TCP	IMAP	Mail retrieval	Plaintext auth risk if not using IMAPS
161/162	UDP	SNMP	Network device management	Default community strings (e.g., 'public') leak info
389	TCP/UDP	LDAP	Directory services	Anonymous binds can leak directory data
443	TCP	HTTPS	Encrypted web traffic	Misconfigured TLS, weak ciphers, cert issues
445	TCP	SMB	Windows file sharing	High-severity exploit target (e.g., EternalBlue/WannaCry)
3306	TCP	MySQL	Database access	Exposed DB, weak/default creds, data breach risk
3389	TCP	RDP	Windows remote desktop	Frequent brute-force & ransomware entry point

Port	Protocol	Service	Common Use	Security Risk
8080	TCP	HTTP (alt/proxy)	Web apps, proxies	Often unmonitored; same risks as port 80 if unencrypted

Port and service data was compiled from standard networking references and IANA-registered port assignments. In line with the task's ethics requirement (own systems only), verification of live/open ports was carried out exclusively on localhost (127.0.0.1) using the command `nmap -sV localhost`. The full scan output is included as an appendix to this report.

Work Done

- Researched 20 commonly used TCP/UDP ports and identified the standard service associated with each.
- Documented the common use case and primary security risk for every port in a structured table.
- Installed and configured nmap on a local macOS system, resolving a Homebrew directory-permission issue encountered during setup.
- Ran an `nmap -sV localhost` scan restricted strictly to the local machine (127.0.0.1), per the task's ethics guidelines.
- Interpreted the scan output, identifying two open ports (5000 and 7000) as Apple AirTunes/AirPlay services.
- Compiled all findings, the reference table, and scan evidence into a single, submission-ready PDF document.

Appendix: nmap Scan Evidence (localhost only)

Command run: `nmap -sV localhost` — scan performed exclusively against 127.0.0.1 (own machine), per task ethics rules.

```
Nmap version 7.99 ( https://nmap.org )
[Platform: x86_64-apple-darwin23.6.0
Compiled with: nmap-liblua-5.4.8 openssl-3.6.3 libssh2-1.11.1 libz-1.2.12 libpcap-1.10.4 nmap-libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: kqueue poll select
192:~ abc$ nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-05 16:06 +0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000037s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
5000/tcp  open  rtsp    rtsp
7000/tcp  open  rtsp    rtsp
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port5000-TCP:V=7.99%I=7%0=7/5%Time=6A4A3AA9P=x86_64-apple-darwin23.6.0
SF:R(GetRequest,8E,"HTTP/1.1\x20403\x20Forbidden\r\nContent-Length:\x200
SF:r\nServer:\x20AirTunes/870.14.1\r\nX-Apple-ProcessingTime:\x200\r\nX
SF:-Apple-RequestReceivedTimestamp:\x201418801\r\n\r\n")%R(RTSPRequest,8E,
SF:"RTSP/1.0\x20403\x20Forbidden\r\nContent-Length:\x200\r\nServer:\x20A1
SF:rTunes/870.14.1\r\nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestRec
SF:ievedTimestamp:\x201418801\r\n\r\n")%R(HTTPOptions,8E,"HTTP/1.1\x20403
SF:\x20Forbidden\r\nContent-Length:\x200\r\nServer:\x20AirTunes/870.14.1
SF:r\nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x
SF:201423818\r\n\r\n")%R(FourOhFourRequest,8E,"HTTP/1.1\x20403\x20Forbid
SF:den\r\nContent-Length:\x200\r\nServer:\x20AirTunes/870.14.1\r\nX-Apple
SF:-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x201423821\r
SF:\n\r\n")%R(SIPOptions,A0,"RTSP/1.0\x20403\x20Forbidden\r\nContent-Leng
SF:th:\x200\r\nServer:\x20AirTunes/870.14.1\r\nCSeq:\x2042\x20OPTIONS\r
SF:nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x201
SF:423824\r\n\r\n");
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port7000-TCP:V=7.99%I=7%0=7/5%Time=6A4A3AA9P=x86_64-apple-darwin23.6.0
SF:R(RTSPRequest,8E,"RTSP/1.0\x20403\x20Forbidden\r\nContent-Length:\x20
SF:0\r\nServer:\x20AirTunes/870.14.1\r\nX-Apple-ProcessingTime:\x202\r\n
SF:X-Apple-RequestReceivedTimestamp:\x201418798\r\n\r\n")%R(GetRequest,8E,
SF:"HTTP/1.1\x20403\x20Forbidden\r\nContent-Length:\x200\r\nServer:\x20A1
SF:rTunes/870.14.1\r\nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestRec
SF:ievedTimestamp:\x201423804\r\n\r\n")%R(HTTPOptions,8E,"HTTP/1.1\x20403
SF:\x20Forbidden\r\nContent-Length:\x200\r\nServer:\x20AirTunes/870.14.1
SF:r\nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x
SF:201423809\r\n\r\n")%R(FourOhFourRequest,8E,"HTTP/1.1\x20403\x20Forbid
SF:den\r\nContent-Length:\x200\r\nServer:\x20AirTunes/870.14.1\r\nX-Apple
SF:-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x201423813\r
SF:\n\r\n")%R(SIPOptions,A0,"RTSP/1.0\x20403\x20Forbidden\r\nContent-Leng
SF:th:\x200\r\nServer:\x20AirTunes/870.14.1\r\nCSeq:\x2042\x20OPTIONS\r
SF:nX-Apple-ProcessingTime:\x200\r\nX-Apple-RequestReceivedTimestamp:\x201
SF:423816\r\n\r\n");
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.20 seconds
192:~ abc$
```

Findings: 998 of 1000 common ports were closed. Two ports were open — 5000/tcp and 7000/tcp — both identified as Apple AirTunes/AirPlay services (Server header: AirTunes/870.14.1) used for AirPlay audio/video streaming between Apple devices. This is a normal, expected finding on macOS and not indicative of a security issue on this host.

Challenges Faced

- Encountered a 'command not found' error when first attempting to run nmap, as it was not yet installed on the system.
- Ran into a Homebrew permissions error during installation, which required changing ownership and write permissions on a system directory before nmap could be installed successfully.
- Needed to distinguish genuinely risky open ports from benign, expected macOS services (AirPlay/AirTunes) when interpreting scan results.

Key Learnings

- Gained a clearer understanding of how common ports map to real-world services and why certain ports are frequently targeted in attacks.
- Learned to use nmap for basic service/version detection and how to read its scan output.
- Understood the practical importance of restricting scanning activity to one's own systems, reinforcing safe and ethical use of networking tools.
- Learned that not every open port indicates a vulnerability — context (e.g., known OS services like AirPlay) matters when assessing risk.
- Practiced troubleshooting real installation/permission issues independently, a skill directly transferable to future technical tasks.

Conclusion

Task 3 provided practical, hands-on exposure to core networking and security concepts within the M-Tech Cyber Security Internship Program 2026. By researching common ports, their services, and associated risks, and by verifying findings through a real nmap scan on a local system, this task strengthened both theoretical understanding and applied technical skill.

The resulting cheat-sheet serves as a practical quick-reference resource, while the troubleshooting encountered during tool setup added further real-world problem-solving experience. This task lays useful groundwork for subsequent, more advanced Networking tasks in the internship roadmap.

Intern Signature:

Name: Ayesha Siddiqua

Date: 5th July, 2026

Supervisor:

Name: Mohazin Zahid

Designation: Supervisor